Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **Python** services, **agentic AI** workflows, distributed data pipelines, and database systems, with strengths in SQL transformation, API design, async processing, fault tolerance, validation, and performance optimization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Operating Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Data Retrieval, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Restructured fault-tolerant **asynchronous agent orchestration** with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at triple throughput without cascading failures under concurrent production workloads.
- Built SQL transformations and incremental integrations across PostgreSQL, SQL Server, and cloud storage with reconciliation and validation, sustaining 5K+ monthly jobs at 99%+ reliability while improving **query performance** 35%.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Built configurable **AI agent workflows** across GPT-4o, Claude, and Qwen-VL with OCR fallbacks, output validation, and concurrent execution, reliably serving 150+ instructors and 10K+ daily production requests.
- Developed batch-processing pipelines for 1K+ uploaded files per run with multithreading, error isolation, and provider routing, maintaining 90%+ **correctness** while reducing feedback turnaround 60% during peak workloads.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*
- Built reproducible Python pipelines processing 100K+ autonomous-driving trajectory logs into training and evaluation features, improving long-horizon planning 18% while benchmarking **large-scale data processing** across diverse simulated workloads.
- Evaluated a spatial-temporal attention encoder against multiple baselines through closed-loop simulation and automated measurements, improving planning stability 15% while systematically documenting **model failure modes** and performance tradeoffs.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Redesigned **ETL pipelines** on AWS Glue and S3, transforming JSON, CSV, and Parquet data from multiple sources into normalized datasets, improving pipeline efficiency 68% and reducing storage costs 15%.
- Optimized backend integrations through object-oriented API design, pagination, caching, and field filtering, improving response times 30% while automating 20+ **unit and integration tests** through nightly Jenkins CI/CD.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Built hourly **Spark micro-batch pipelines** on Databricks over 50K+ booking and payment events, producing validated aggregates for peak demand, active users, and revenue while supporting real-time operational reporting.
- Developed Python APIs with idempotent payment webhooks, transactional workflows, and Firebase synchronization, maintaining sub-200ms responses while reducing manual processing 70% through **reliable event-driven integration** across distributed clients.

Skills

Languages: Python, Java, C++, SQL, TypeScript, JavaScript, T-SQL
AI & Backend: Agentic AI, LLM APIs, Claude, GPT-4o, Prompt Engineering, FastAPI, REST APIs, API Design, Async Processing
Databases: Snowflake, PostgreSQL, SQL Server, Oracle, MySQL, SQLite, Query Optimization, Schema Design, Stored Procedures
Data Systems: ETL/ELT, Spark, Databricks, AWS Glue, Incremental Loads, Data Validation, Micro-Batching, Distributed Processing, Fault Tolerance
Cloud & Quality: AWS, Azure, GCP, Docker, Jenkins, CI/CD, PyTest, Unit Testing, Integration Testing, Benchmarking, Monitoring

Projects

SchemaShift | *Python, FastAPI, SQL AST, Snowflake, Spark, PyTest, Docker*
- Designed an **agentic code-conversion pipeline** that translates legacy SQL and stored procedures, generates synthetic fixtures, executes source-target comparisons, and routes semantic discrepancies through repair and repeatable validation workflows.

Chat2SQL Code | *Java, TypeScript, Snowflake, SQL Server, Oracle, Docker*
- Extended an AI-assisted database workspace supporting 10+ engines, including **Snowflake, SQL Server, and Oracle**, with schema synchronization and validated query execution while achieving 82% accuracy on generated SQL.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP, REST APIs*
- Built a **retrieval and evaluation pipeline** using LLaMA-2, transformer embeddings, and FAISS over 50K+ documents, reducing retrieval latency 40% while sustaining 95% query accuracy for 200+ users.